Financial Document QA on FinanceBench

Live demo: finqa-demo.vercel.app (step-by-step replay on sample questions, and every answer with its cited pages). Code and full results: github.com/moriowen/finqa. The system diagram is on page 3.

This is a comparative experiment, not a product. Qwen3.5 at 9B and 27B (and 2B in the first experiment) were compared with and without a checking pipeline, to decide which model and setup to use later. One rule shaped the design: a wrong answer is worse than no answer. If a configuration cannot point to evidence that code has checked, it refuses and gives the reason. Results are reported as three counts out of 150 questions: correct, wrong and refused.

Terms used

Term	Meaning
Bare	The model gets the question and the retrieved pages in one prompt and answers. Nothing checks it.
Full pipeline	The same model and pages inside the checking pipeline of section 1. Any failed check ends in a refusal.
Refused	No answer is given (NOT_FOUND or AMBIGUOUS), with the reason. Counted apart from wrong answers.
Provenance	The filing, PDF page, exact quote and surrounding passage behind each value in an answer.
Router, reader	Router: the LLM call that picks the filing when none is named. Reader: the calls that read the pages and answer.
Statement pack	The primary statement pages of a filing: income statement, balance sheet, cash flow statement.
BM25, embedding search	BM25 ranks pages by the question's words, rare words counting more. Embedding search ranks them by similarity of meaning.
Gold answer, gold page	The answer and evidence page written by FinanceBench's analysts. Used only for grading.

1. System design

Parsing. PyMuPDF reads each of the 363 filings and stores every page twice: plain text for search, and layout text that keeps table rows and columns for the model to read. One chunk is one PDF page, so tables stay whole and the page number is the provenance. Pages whose title is a primary statement form the statement pack, because most gold evidence is on them.

Retrieval. Code, not the model, picks about 12 pages from the filing: the statement pack plus the best pages from BM25 and from embedding search (Qwen3-Embedding-8B), merged by rank. This finds a gold page for 87% of the dev questions. When no filing is named, a router call first names the company, year and type, and code narrows the 363 filings to up to three candidates.

Generation. Three LLM calls, with Python checks between them. (1) The reader returns JSON, with the schema enforced by vLLM: each value with page, exact quote, column and period, plus a formula over value names. (2) Python checks that each quote is on its page and holds the value, checks the formula, and does the arithmetic. (3) The reader writes the answer from the checked values only, and Python checks every number in it. (4) A verifier from another model family (Gemma 4 12B) checks period, unit, column and support on images of the cited pages. Code then attaches the provenance. A failed check gets one retry, then a refusal. Candidate filings that disagree return AMBIGUOUS.

The pipeline was tuned on 51 dev questions, and the other 99 were run once at the end. Numeric answers were graded by code, text answers by gpt-oss-20b, which agreed with the FinanceBench paper's 2,400 human labels 77% of the time.

2. Answers with provenance

Every answer and refusal of every configuration, with page, quote, passage and an image of the page, is in the demo and the repository's results folder.

Experiment 1: the filing is given (tests reading)

Configuration	Correct	Wrong	Refused
9B, bare	93	57	0
9B, full pipeline	72	16	62
27B (FP8), bare	101	48	1
27B (FP8), full pipeline	80	20	50
GPT-4 Turbo, paper, single store (human labels)	75	17	58

2B answered only 14 questions with the full pipeline (12 correct) and was not used further. On 290 extra questions whose filing holds no answer, the full pipeline gave 0 false answers with 9B and 1 with 27B. The same models without checks gave 20 and 22.

Experiments 2 and 3: the system must find the filing (tests finding and reading, and which of the two needs the larger model)

Finds the filing	Reads	Correct	Wrong	Refused
9B	9B	59	20	71
27B	9B	58	20	72
27B	27B	76	16	58
GPT-4 Turbo, paper, shared store (human labels)		29	20	101

Two sample answers from 9B, and one refusal:

- *Which of JPM's segments had the highest net income in 2022 Q2?* Corporate & Investment Bank, $3,725 million, the max of four segment values. Evidence: JPMORGAN_2022Q2_10Q, page 27, "Net income | $3,725", with pages 22, 33 and 37 for the other segments.
- *Does American Water Works have positive working capital in FY2022?* No: $1,250 million minus $2,811 million is -$1,561 million. Evidence: AMERICANWATERWORKS_2022_10K, pages 81 and 82.
- *Is Verizon's liquidity healthy based on its quick ratio for FY2022?* Refused: the inputs were found on page 56, but not which items the quick ratio should include.

3. What worked, what didn't, and what would change

What worked. The checks did more than model size to prevent wrong answers: they cut wrong answers by more than half for each model (9B: 57 to 16, 27B: 48 to 20) and almost removed false answers on questions with no answer, at the cost of turning some correct answers into refusals. Letting code do the arithmetic and attach the citations means no answer can cite text that is not on its page. Finding the filing works with a small model: the 9B and 27B routers rank the right filing first equally often (119 and 118 of 150), and swapping the router made no difference to the answers. Most router misses come from questions that do not say which document they mean, which calls for asking the user, not for a larger model.

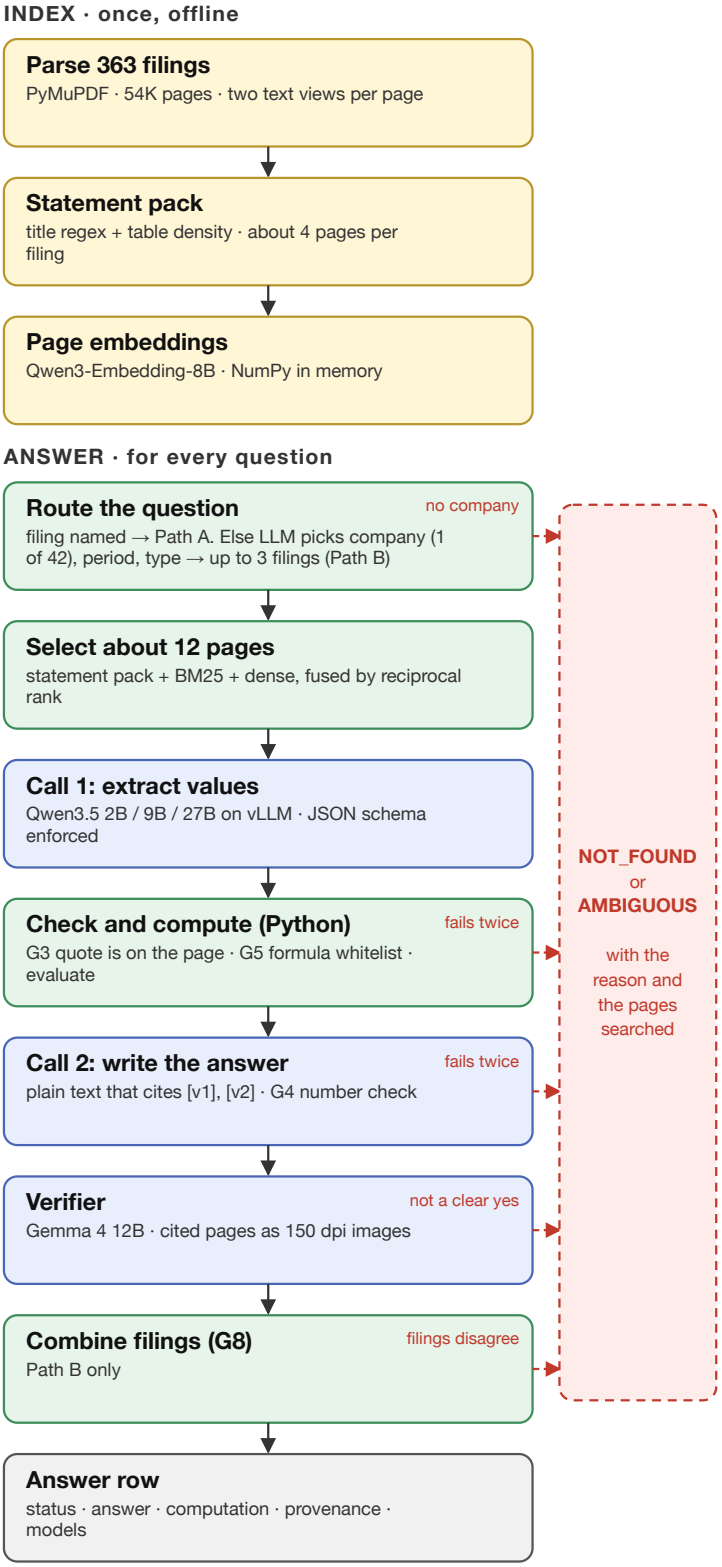
What didn't. Reading is where size matters. With the same router, the 27B reader got 18 more correct answers than 9B and 4 fewer wrong ones, so finding the filing cost 9B 13 correct answers but 27B only 4. With the filing given, 27B gets more correct answers but also 4 more wrong ones than 9B, because it passes the code checks more often and the verifier catches only about a quarter of the wrong answers that reach it. Most remaining errors use a real, correctly cited number for a different definition of the metric, such as working capital from total instead of operating items, which no check catches. Speed: one question takes 18 s with 9B and 48 s with 27B, and almost all of it is the first call writing its JSON. Retrieval and the checks take milliseconds.

What would change. A small model to find the filing and the larger model to read it. A verifier that also checks the formula against the question, and answers that state which definition of a metric they used. Searching the best-ranked filing first and stopping when it answers, instead of reading every candidate. A shorter first-call output and fewer retries to cut the wait. Limited compute meant 27B in FP8, a 12B verifier, a 20B judge and one run each. More compute would allow larger models, verifier and judge, and repeated runs.

How a question becomes an answer with provenance

Left: each stage and its technology. Right: the decision made at that stage and why. Red: where the system refuses.

data / index · code (deterministic) · LLM call · output · refusal path



DECISION AND REASON

Chunk = one PDF page, not a token window. Tables stay whole, and the page number is the provenance. Plain text feeds search. Layout text (columns joined by "|") is what the model reads.

Most gold evidence is on the income statement, balance sheet or cash flow statement, so these pages are always shown to the model.

No vector database: 54K pages fit in memory. BM25 is built per filing at question time (largest filing: 667 pages).

DECISION AND REASON

Path B keeps up to 3 candidate filings instead of guessing one. Each runs alone, so two filings never share a prompt.

Picked on dev with a rule fixed in advance: the cheapest option within 5 points of the best. Gold-page recall 84% → 87% with dense. LLM query planning added cost, not recall.

Each value comes with its page, exact quote, column, period and scale, plus a formula over value names. The model picks the numbers. It never does arithmetic.

Formulas may use only value names and avg, sum, cagr, abs, max, min. Constants only if they appear in the question. A failed check gets one retry with the error.

Every number in the text must be a checked value or the computed result. One retry with the error.

A different model family, so it does not share the answering model's mistakes. Checks period, unit, column and support. Anything but a clear yes is a refusal.

One filing answers → that answer. All agree → answer. They disagree → AMBIGUOUS, with each value and its source.

Code attaches the provenance from checked values: filing, PDF page, quote, passage, column, scale. The LLM never writes a page number. No checked citation, no answer.

Evaluation sits outside the system and never feeds it: code grades numeric answers and gpt-oss-20b grades text answers against the gold answer. 51 dev questions for every choice above, 99 test questions run once.